Queen Mary
University of London
Science and Engineering

EBU4202: Digital Circuit Design
Karnaugh Maps

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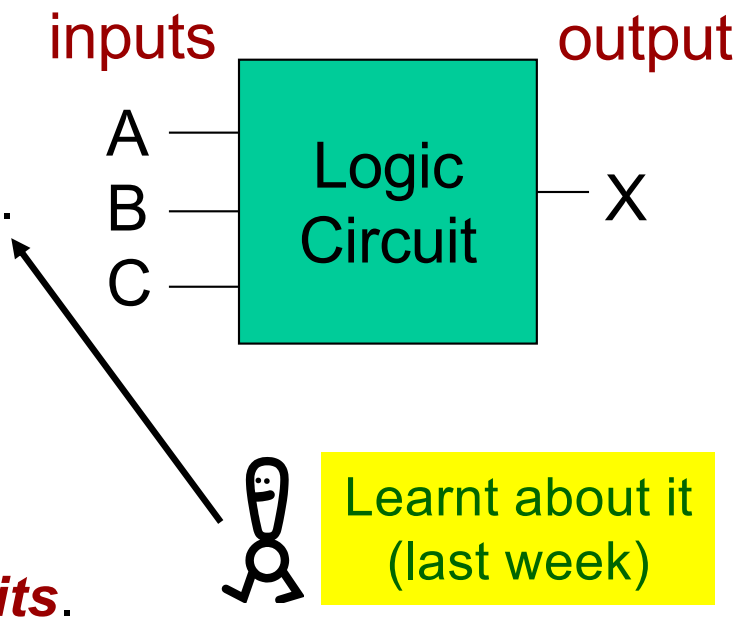
Recap: Logic Circuits – Analysis & Synthesis

- **Digital Circuits**:
  - **Combinational Logic Circuit**: a circuit whose outputs depend on its current inputs.
  - **Sequential Logic Circuit**: a circuit whose outputs depend not only on current inputs, but also on past inputs.



- These slides are concerned with the ***analysis*** and ***synthesis*** of ***combinational*** logical ***circuits***.
  - **Analysis** → start from a logic diagram of a circuit and derive a formal description of the function of the circuit.
  - **Synthesis** → start with a formal description of the function of a circuit and proceed to a logic diagram that performs the required function.

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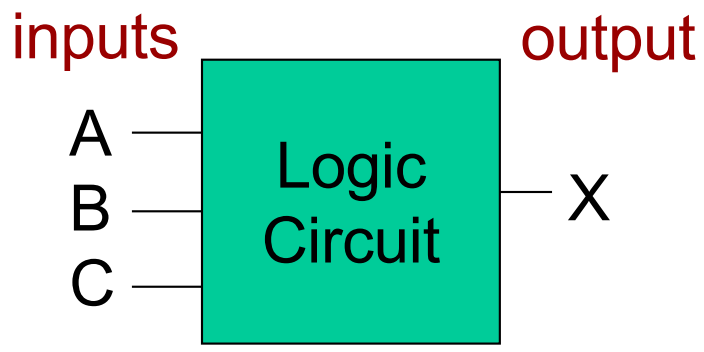

Studied later in the course.

- These slides are concerned with the ***analysis*** and ***synthesis*** of ***combinational*** logical ***circuits***.
  - **Analysis** → start from a logic diagram of a circuit and derive a formal description of the function of the circuit.
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Overview: Switching Algebra & Combinational Logic Design

* **Switching Algebra**
* **Combinational Circuit Analysis & Synthesis**

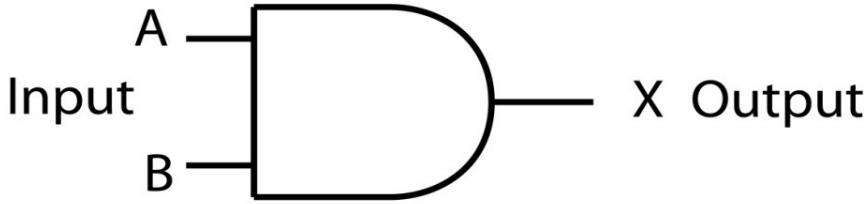
Chapters 4 & 6 – "Digital Design: Principles and Practices" book

Combinational Circuit Analysis

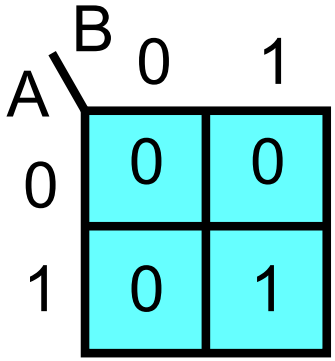
- ***Analysis of combinational circuits***:
  - It requires a formal description of their logic function.
  - *Logic function description* allows:
    - *Determination* of circuit behaviour for different input combinations.
    - *Manipulation* of an algebraic description to derive different circuit structures for the logic function.
    - *Transformation* of an algebraic description into a form corresponding to an available circuit structure.

Karnaugh Maps

- *Karnaugh Map*:
  - Another approach to represent (and simplify) Boolean equations.
  - **Example** (Karnaugh map for a 2-input AND gate):



Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1



Karnaugh Map

Karnaugh Map Simplification

- **Why use Karnaugh map simplification**:
  - Working with algebraic equations is often tedious and error-prone.
  - A mapping technique can be used to reduce an algebraic equation to its simplest form.
  - Simplification approaches used (consisting of grouping adjacent *1s* or *0s* in *powers of 2*) are *minterm* or *maxterm*.
  - Equations of up to 5 variables can be easily simplified by hand. However, it becomes complicated to simplify equations with more than 5 variables.

2-Variable Map (1/2)

- How to **build a 2-variable Karnaugh map**:
  - There will be 4 *minterms* for a 2-variable equation, so use a Karnaugh map with 4 squares (i.e., a 2x2 table).
  - *1's* and *0's* on the left side and top of the map designate the values of the variable.
  - Variable *X* is ***complemented*** in row *0* and ***uncomplemented*** in row *1*.
  - Variable *Y* appears ***complemented*** in column *0* and ***uncomplemented*** in column *1*.

X \ Y	0	1
0	m0	m1
1	m2	m3

X	Y	Product Term	Symbol
0	0	$\overline{X}\cdot\overline{Y}$	m_0
0	1	$\overline{X}\cdot Y$	m_1
1	0	$X\cdot\overline{Y}$	m_2
1	1	$X\cdot Y$	m_3

2-Variable Map (2/2)

- How **a 2-variable Karnaugh map works**:
  - The concept is to put a *1* in each square that has a corresponding *minterm* in the Boolean equation.
  - Only four terms are possible in *Sum of Products* form.

X \ Y	0	1
0	m0	m1
1	m2	m3

Equates to

X \ Y	0	1
0	$\overline{X}\bullet\overline{Y}$	$\overline{X}\bullet Y$
1	$X\bullet\overline{Y}$	$X\bullet Y$

X	Y	Product Term	Symbol
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1	0	$X\bullet\overline{Y}$	m_2
1	1	$X\bullet Y$	m_3

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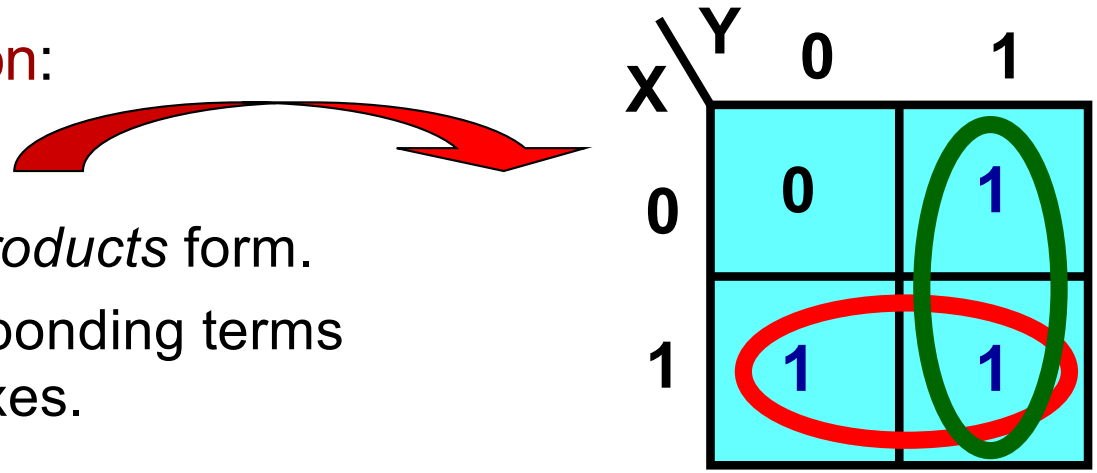
Example: 2-Variable Map

- Simplify the boolean equation:

  F = X'•Y + X•Y' + X•Y

  – Function ***F*** is in *Sum of Products* form.

  – Put *1's* in boxes of corresponding terms and put *0's* in all other boxes.



- Simplified expression for ***F*** is obtained by **grouping adjacent 1's** (in <u>powers of 2</u>) and eliminating unnecessary variables.

- Here, **there are two ways of grouping *1's***: one way is to circle the row where *X=1*; the other way is to circle the column where *Y=1*.

- ***Y*** is eliminated from the two product terms where *X=1* and ***X*** is eliminated from the two product terms where *Y=1*.

***Answer*: F = X + Y.**

Additional slide

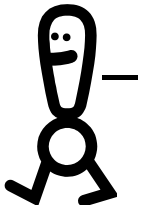
Remember to circle the **largest groupings possible!**

Golden Rules:

1. No zeroes allowed.
2. No diagonal groupings
3. Only power of 2 cells in each grouping
4. Every 1 must be at least in one grouping
5. Overlapping is allowed
6. Groups may “Wrap-around”
7. Fewest number of groups possible.

3-Variable Map (1/2)

- How to **build a 3-variable Karnaugh map**:
  - There will be 8 *minterms* for a 3-variable equation, so use a Karnaugh map with 8 squares (usually, a 2x4 table).
  - The *minterm* numbers do not follow the normal binary counting order sequence.
  - Only 1 bit changes from one adjacent column to the next.



What does this remind you of?

X \ YZ	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

Y

Z

3-Variable Map (2/2)

- How **a 3-variable Karnaugh map works**:

X \ YZ	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

Equates to

X \ YZ	00	01	11	10
0	$\bar{X}\bullet\bar{Y}\bullet\bar{Z}$	$\bar{X}\bullet\bar{Y}\bullet Z$	$\bar{X}\bullet Y\bullet Z$	$\bar{X}\bullet Y\bullet\bar{Z}$
1	$X\bullet\bar{Y}\bullet\bar{Z}$	$X\bullet\bar{Y}\bullet Z$	$X\bullet Y\bullet Z$	$X\bullet Y\bullet\bar{Z}$

Example 1: 3-Variable Map

Simplify: $F(X, Y, Z) = \Sigma m(1, 2, 5, 6)$ **= 001, 010, 101, 110**

Note: Circle *1's* in horizontal or vertical groups of 1, 2, 4 or 8 only (*powers of 2*)!

or

$$= \bar{X}\bar{Y}Z \quad \bar{X}Y\bar{Z} \quad X\bar{Y}Z \quad XY\bar{Z}$$

$$m1 + m5 = \bar{X}\bullet\bar{Y}\bullet Z + X\bullet\bar{Y}\bullet Z$$

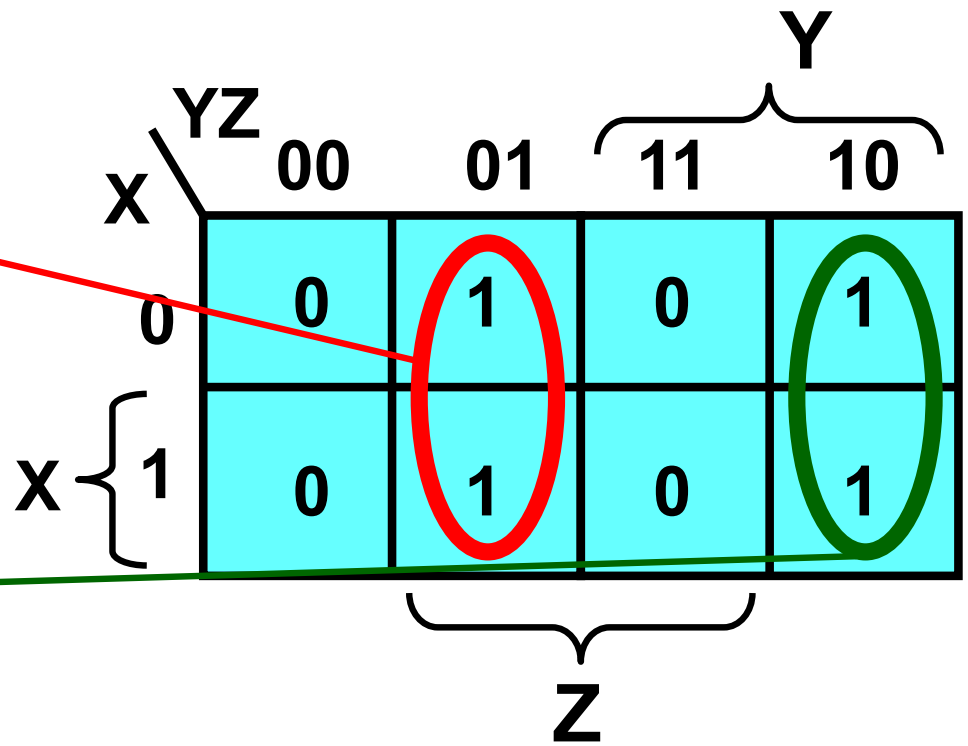
$$= (\bar{X} + X)(\bar{Y}\bullet Z)$$

∴ X is redundant

$$m2 + m6 = \bar{X}\bullet Y\bullet\bar{Z} + X\bullet Y\bullet\bar{Z}$$

$$= (\bar{X} + X)(Y\bullet\bar{Z})$$

∴ X is redundant



ANSWER: F = Y'Z + YZ'

Example 2: 3-Variable Map

To be completed in class …

Simplify, using a Karnaugh map:

$F(X, Y, Z) = \Sigma m(3, 4, 5, 6, 7)$

X \ YZ	00	01	11	10
0				
1				

Check using Boolean Algebra

$F = X'YZ + XY'Z' + XY'Z + XYZ' + XYZ$

$= XZ(Y' + Y) + XZ'(Y' + Y) + X'YZ$

$= XZ + XZ' + X'YZ$

$= X(Z + Z') + X'YZ$

$= X + X'YZ$

$= X + YZ$ [since $X + X'Y = X + Y$]

Example

$F(X, Y, Z) = \Sigma m(3, 4, 5, 6, 7)$

X \ YZ	00	01	11	10
0			1	
1	1	1	1	1

YZ

X

Check using Boolean Algebra

$$
\begin{aligned}
F &= X'YZ + XY'Z' + XY'Z + XYZ' + XYZ \\
&= XZ(Y' + Y) + XZ'(Y' + Y) + X'YZ \\
&= XZ + XZ' + X'YZ \\
&= X(Z + Z') + X'YZ \\
&= X + X'YZ \\
&= X + YZ \quad [\text{since } X + X'Y = X + Y]
\end{aligned}
$$

4-Variable Map

WX \ YZ	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

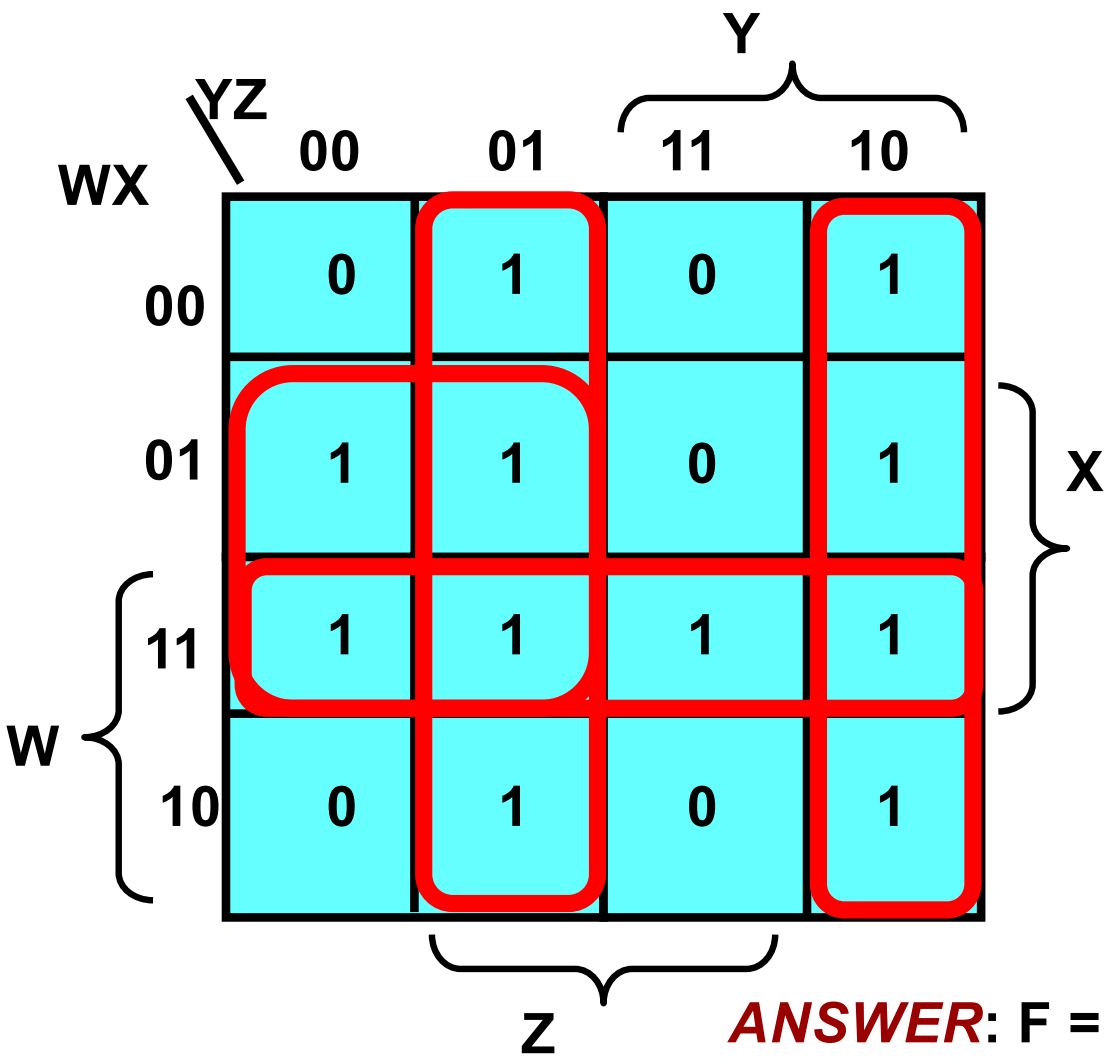
Y

X

W

Z

Example 1: 4-Variable Map



W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Truth Table

ANSWER: $F = X\overline{Y} + \overline{Y}Z + Y\overline{Z} + WX$

Example 2: 4-Variable Map

- This is an example of a Karnaugh map with all possible groupings (except circles with a single value):

F = W'Y' + WXZ + WX'Y.

WX \ YZ	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	1	1	0
10	0	0	1	1

Prime Implicants

- **Challenge when using K-maps**: To select the right groups.
  - **IF** the number of groups is not minimised **AND**
  - the size of each group is not maximised **THEN**,
    - Resulting expression will still be equivalent to the original one. **BUT**
    - Resulting expression will not be a ***minimal*** sum of products (or MSP).

- Good **approach to finding an actual MSP**:
  - Find all of the largest possible groupings of *1's* (these are called the ***prime implicants***); but only in <u>powers of 2</u>!
  - The final MSP will contain a subset of these prime implicants.

Example: K-Map with Prime Implicants

- Here is an example of a Karnaugh map with *prime implicants* (all of the marked groups are prime implicants):

almost

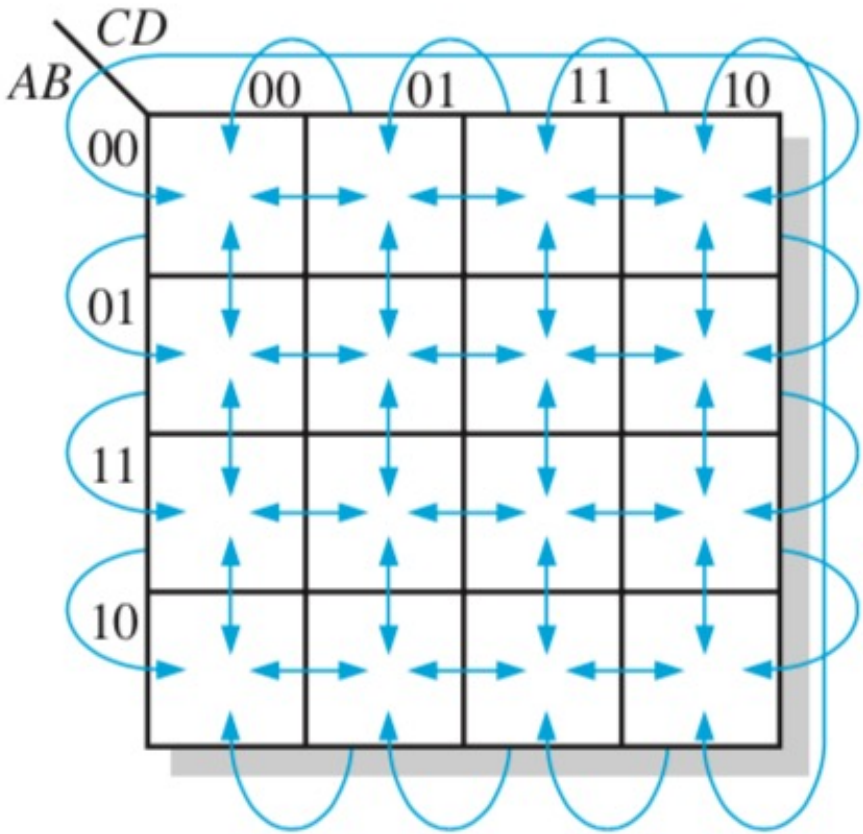
WX \ YZ	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	1	1	0
10	0	0	1	1

Why is this not a prime implicant?

Additional: K-Map with Prime Implicants

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

Cell Adjacency



Wraparound

Additional: K-Map with Prime Implicants – Wrap-around Demo

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

Additional: K-Map with Prime Implicants

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

Good approach: Use the fewest possible number of maximal groupings needed to cover all of the squares marked with a 1.

$$F(A, B, C, D) = BD + CD + AC + B'D'$$

Essential Prime Implicants (EPIs)

- If any group contains a ***minterm*** that isn't also covered by another overlapping group, then that is an ***EPI***.
- ***EPIs*** appear in the MSP, since they contain ***minterms*** that no other groups include.
- **Example**:

EPI: because no other group covers *m0*, *m1*, and *m4*.

EPI: because no other group covers *m10*.

WX \ YZ	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	1	1	0
10	0	0	1	1

Another Example : Covering the other *Minterms*

- Example from previous slide:
  - Pick as few other prime implicants as necessary, to ensure that all the *minterms* are covered.
  - After choosing the **green** rectangles in the example, there are just two *minterms* left to be covered: *m13* and *m15*.
  - These are both included in the **red** prime implicant, WXZ.
  - The resulting equation is,

$$F = W'Y' + WXZ + WX'Y$$

WX \ YZ	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	1	1	0
10	0	0	1	1

Additional: Essential Prime Implicants

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	0	1	1
10	0	0	1	0

Additional: Essential Prime Implicants

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	0	1	1
10	0	0	1	0

F(A, B, C, D) = A'D + CD + BD'

Karnaugh Maps Part 1 Summary

- Build a Karnaugh Map
  - Truth Table
  - Sum of Minterms
- Simplify a Logic Function using its Karnaugh Map
  - Group 1s
  - Identify which variables are redundant
- Obtain the Minimal Sum of Products (MSP)
  - Prime Implicants
  - Essential Prime Implicants